

NIKOLAY TULUPOV

Reinforcement Learning & Control Systems Engineer

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PROFILE

Control systems and reinforcement learning engineer with over two years of engineering and applied R&D experience. Specializes in learning-based control of PMSM electric drives, simulation, experimental validation, and engineering software development using Python, PyTorch, and MATLAB/Simulink. Co-author of an IEEE EDM 2026 publication and two registered software systems.

CORE EXPERTISE

Machine Learning & RL: Python, PyTorch, reinforcement learning, TD3, actor-critic methods, reward design, experiment design

Control & Modeling: control systems, PMSM/IPMSM drives, FOC, mathematical modeling, MATLAB/Simulink, Motor-CAD

Engineering & Tools: C/C++, Git, Linux, CAN, STM32/AT32, power inverters, test-bench experiments

RELEVANT EXPERIENCE

Moscow Institute of Physics and Technology

Mar 2026 – Present

Research Engineer, Robotics and Electric Drive Laboratory

Moscow, Russia

- Research and develop PMSM electric-drive control methods, with a focus on reinforcement learning for closed-loop control of physical systems.
- Developed and evaluated a physics-informed TD3 approach for end-to-end PMSM torque control and compared it with a classical FOC baseline.
- Build software pipelines for test-bench data processing, experiment automation, quantitative evaluation, and visualization.
- Automate Motor-CAD calculations and electric-motor geometry optimization.
- Conduct experimental validation and prepare scientific publications and technical presentations on electric-drive control.

AmpereMagnete

Mar 2025 – Mar 2026

Engineering Intern, Artificial Intelligence Group

Troitsk, Moscow, Russia

- Developed and experimentally evaluated PMSM/IPMSM control and calibration methods, including reinforcement-learning-based approaches.
- Developed embedded firmware for an AT32-based power inverter and software for remote inverter control and test-bench operation.
- Implemented CAN-based telemetry acquisition, visualization, logging, and experimental data analysis.
- Developed OptiRunMotorCAD for automated calculations and motor-geometry optimization; contributed to the registered StandControl application.

EDUCATION

Moscow Institute of Physics and Technology

2026 – 2028 (expected)

M.Sc. in Applied Mathematics and Physics

Moscow, Russia

Specialization: E-Mobility – Technologies, Materials and Control

Moscow Institute of Physics and Technology

2022 – 2026

B.Sc. in Applied Mathematics and Physics

Moscow, Russia

Concentration: Radio Engineering and Computer Technology

ADDITIONAL ENGINEERING EXPERIENCE

TIENDA LLC

Apr 2026 – Present

Engineer, Technical Department

Moscow, Russia

- Continue development of a remote-control system for the Motorola XTL2500 radio following the project's organizational transfer from PromSvyazRadio.

PromSvyazRadio LLC

Feb 2024 – Apr 2026

Assistant to the Chief Engineer, Technical Department

Moscow, Russia

- Developed a distributed client-server system on Linux-based single-board computers for remote radio control, real-time bidirectional audio, and command processing.
- Implemented SB9600 protocol communication, hardware interfaces, and end-to-end hardware/software integration; contributed to the schematic, PCB, enclosure, testing, and technical documentation.

SELECTED RESEARCH & ENGINEERING PROJECTS

Physics-Informed TD3 for End-to-End Torque Control of a PMSM Drive

- Designed the state and action spaces, reward function, curriculum, and evaluation pipeline for a continuous-control TD3 agent.
- Introduced physics-informed critic objectives based on monotonicity constraints, auxiliary next-state prediction, and switch-aware TD weighting.
- Achieved a torque MAE of 0.079 N m and a 98.9% success rate in the final evaluation, with performance close to classical FOC.
- Evaluated robustness to measurement noise and analyzed real-time MCU deployment constraints. Published at IEEE EDM 2026.

OptiRunMotorCAD

Developed a software application for automating Motor-CAD calculations and electric-motor geometry optimization. Registered as a computer program in 2025 ([No. 2025693472](#)).

StandControl

Co-developed a client-server application for remote power-inverter control, CAN telemetry monitoring, visualization, logging, and test-bench operation. Registered as a computer program in 2026 ([No. 2026614166](#)).

SELECTED COURSEWORK & TRAINING

[Reinforcement Learning](#), MIPT (Spring 2026) | [Deep Reinforcement Learning Course](#), Hugging Face

PUBLICATIONS & RECOGNITION

- N. Tulupov and D. Kanurin, "[Physics-Informed TD3 for End-to-End Torque Control of a PMSM Drive](#)," 2026 *IEEE 27th International Conference of Young Professionals in Electron Devices and Materials (EDM)*, 2026. DOI: 10.1109/EDM69524.2026.11632234.
- N. Tulupov, S. Makov, D. Kanurin, and A. Tishin, "[PMSM Electric Drive Calibration Method Using a Reinforcement Learning Algorithm](#)," *XII International Conference Engineering & Telecommunications (En&T 2025)*, MIPT, Moscow, 2025; abstract published in the proceedings.
- N. Tulupov, S. Makov, D. Kanurin, and A. Tishin, "[Calibration Method for a Permanent-Magnet Synchronous Motor \(PMSM\) Using Deep Reinforcement Learning](#)," *III International Youth Conference "AI Technologies in Science and Education"*, Moscow State University, Moscow, 2025; abstract published in the proceedings. Best Presentation Award, Exact and Engineering Sciences section.
- N. Tulupov and D. Kanurin, "[Hybrid Control Architecture for a PMSM Electric Drive Using Reinforcement Learning](#)," *68th All-Russian Scientific Conference of MIPT*, Moscow, 2026; abstract published in the proceedings.

LANGUAGES

Russian – Native | English – B2 (Upper-Intermediate)